

8. Compound **A** contains only carbon, hydrogen and an unknown halogen.

Refluxing compound **A** in aqueous sodium hydroxide followed by the addition of nitric acid and aqueous silver nitrate produces a white precipitate.

Elemental analysis of compound **A** indicates it contains 39.02% carbon and 3.25% hydrogen by mass.

When bromine is added to compound **A**, 123 g of compound **A** reacts with 320 g of bromine.

The ^1H NMR spectrum of compound **A** consists of only one peak. The ^{13}C NMR spectrum of compound **A** consists of two peaks.

The infrared spectrum and simplified mass spectrum are shown below and overpage.

